

Choose the most appropriate answer and fill it in the attached answer sheet.

You may choose answer E (=NONE) if none of the answer fits.

- $E_a = \frac{C \cdot \Phi \cdot Z}{60} \cdot \frac{P}{A}$
- 1) A wave wound DC machine has 400 conductors and 8-poles. The voltage induced per conductor is 1.5-V. The machine will generate a voltage equal
 a) 100-V ☒ b) 300-V c) 400-V d) 800-V
 - 2) A lap wound DC machine has 600 conductors and 6-poles. The total generated voltage is 200 V. The voltage induced per conductor equal
☒ a) 2-V b) 4-V c) 1.5-V d) 3-V
 - 3) A lap wound DC machine has 600 conductors and 4-poles. The resistance per conductor is $5 \times 10^{-4} \Omega$. The total armature resistance equal
 a) 0.253- Ω b) 0.125- Ω ☒ c) 0.01875- Ω d) 0.0375- Ω
 - 4) A 8- poles wave wound DC machine has full load armature current of 50 A. The conductor current equal
 a) 100-A b) 50-A c) 12.5-A ☒ d) 25-A
 - 5) The lap wound DC machine is used for high voltage, and high current
 a) True ☒ b) False
 - 6) For lap wound DC machine, the value of $C/2p$ should be integer.
☒ a) True b) False
 - 7) For a progressive lap wound DC machine, the commutator pitch equal to
 a) $C/2P$ b) C/P ☒ c) 1 d) -1
 - 8) For a progressive wave wound DC machine, the commutator pitch equal to
 a) $(C+1)/2P$ b) $(C-1)/P$ ☒ c) $(C+1)/P$ d) $C/2P$
 - 9) Commutator in DC generator is used to
 a) collect the current b) reduce losses ☒ c) convert AC armature voltage to DC d) increase efficiency

Questions (10-12)

A 6-pole dc machine generator has a lap-connected armature with 120 slots and 8 conductors per slot.

- $Z = 8 \times 120$
 $C = \frac{Z \cdot N \cdot P}{60}$
 $G_a I_a$
- 10) The total flux per pole of the generator if it rotates at 500-rpm and the generator emf is 240-V, is ..
 a) 0.015-Wb b) 0.06-Wb c) 0.045-Wb ☒ d) 0.03-Wb
 - 11) The developed power if the armature current is 30-A is
 a) 14.4-kW b) 8.2-kW ☒ c) 7.2-kW d) 7.5-kW
 - 12) The peak amplitude of armature reaction, if the armature current is 30-A, is
 $\frac{1}{2} \frac{Z I_a}{P A}$

a) 1800-AT/pole

b) 400-AT/pole

c) 1200-AT/pole

d) 900-AT/pole

13) The series field of a long-shunt compound dc motor is excited by

a) supply current

b) separate current

c) shunt field current

d) armature current

14) The necessary conditions for operating DC machine as shunt generator,

a) There should be residual flux

b) Field resistance should be higher than critical value

c) The main flux should aid the residual flux

d) a & c

15) In a DC machine, without brush shifting, the peak value of armature reaction Ampere-turn at the direction of

a) GNA

b) MNA

c) Direct Axis

d) a & b

16) In DC generator, due to the armature reaction, the MNA is shifted opposite to the direction of rotation

a) True

b) False

17) The emf induced in a conductor of a DC machine driven at 1000-rpm, the peak value of flux density is 1.0-Tesla, diameter of armature 2.0-m, and length of machine equals 0.3-m is

a) 41.83-V

b) 29.58-V

c) 31.42-V

d) 18.85-V

18) For 6000 maximum armature reaction Ampere-turns per pole, the value of armature reaction Ampere-turns at an angle of 30° mechanical from the geometric neutral axis (GNA) for a 6-pole DC machine equal

a) 521

b) 0

c) 5231

d) 571

Question (19,20)

19) For 6000 maximum armature reaction Ampere-turns per pole, if brush shift is 2° mechanical, The demagnetizing Ampere-turn per pole for a 6-pole DC machine will be

a) 400

b) 5835.2

c) 416.8

d) 571

20) The cross-magnetizing Ampere-turn per pole will be

a) 521

b) 416.8

c) 5835.2

d) 5600

21) The number of conductors on each pole-shoe required in a compensating winding for a 6-pole wave-wound DC armature containing 288 conductors, assume ratio of (pole-arc/pole-pitch = 0.75), is

a) 6

b) 8

c) 9

d) 18

Question (22-24)

A dc shunt generator has the following magnetization curve at 1000-rpm,

I_f (A)	0	0.1	0.2	0.3	0.4	0.5	0.75	1.0	1.5
E_b (V)	4	32	64	96	120	136	165	178	196

22) If the no load voltage at 1000-rpm is 165-V, the shunt field resistance R_f is

a) 165-Ω

b) 178-Ω

c) 200-Ω

d) 220-Ω

23) The minimum running speed at this R_f is

a) 687.5 rpm

b) 823.5 rpm

c) 115 rpm

d) 600 rpm

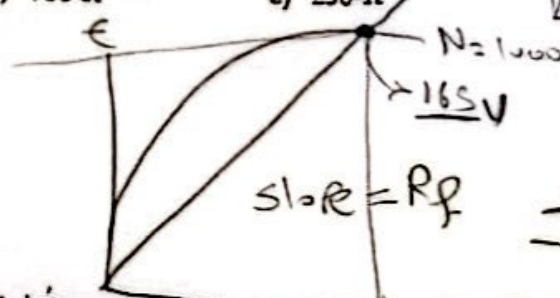
24) The critical field resistance of this generator when running at 600-rpm is.

a) 320-Ω

b) 160-Ω

c) 256-Ω

d) 192-Ω



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$$R_f = 320 = \frac{E_c}{0.3}$$

$$\frac{R_{fc1}}{R_{fc2}} = \frac{N_1}{N_2}$$

25) Armature voltage control for separately excited DC motor is used to control the speed above the rated speed

a) true

~~b) false~~

26) Ward Leonard control technique is basically....

a) An armature voltage control method

b) Field control method

c) An armature resistance control method

~~d) Both a & b~~

27) A DC shunt generator delivers a load current of 50-A at 200-V. The armature and shunt field winding resistances are respectively, $R_a = 0.2\text{-}\Omega$ and $R_f = 100\text{-}\Omega$. The rotational losses are 400-W. The efficiency of the generator in this case is (neglect armature reaction)

a) $\eta_{FL} = 90\%$

~~b) $\eta_{FL} = 88.18\%$~~

c) $\eta_{FL} = 84\%$

d) $\eta_{FL} = 88.96\%$

Question (28,29)

A 250 -V series dc motor with compensating winding, and a total series resistance $R_a + R_s$ of $0.08\text{ }\Omega$. The series field consists of 25 turns per pole. With the following magnetization curve at 1200-rpm.

I_f	0	16.66	33.33	50	64	68	72	76	80
E_a (V)	4	32	64	96	120	136	165	178	196

28) For an armature current of 50 A, the rotor speed will be

~~a) 3075 rpm~~

b) 2500 rpm

c) 1225 rpm

d) 1085 rpm

29) While the induced torque is equal to

a) 20.1 Nm

~~b) 38.2 Nm~~

c) 50 Nm

d) 60.3 Nm

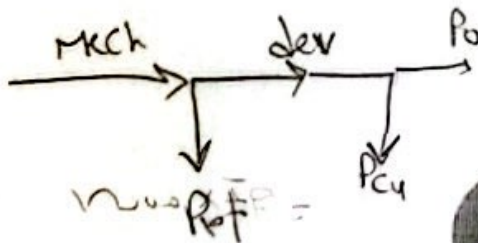
30) For a differential long compound motor, if the shunt field circuit is opened

a) The machine will continue running in the same direction with high speed

b) The speed of the machine will decrease till it rotates in the opposite direction with high speed

c) The speed of the motor will not change

~~d) The machine will continue running in the same direction with low speed~~



With best wishes

Ass.Prof. M. Moustafa

$R_{ex} \uparrow \quad N \downarrow$

Speed Control

arm Control

Field Control

$\rightarrow Q \downarrow \quad \uparrow N$

$$\omega = \frac{V_t - I_a R_a}{K \Phi}$$

Separate excited motor $\rightarrow V_t \uparrow \quad \omega \uparrow$

Solution

1) $Z = 400$ $Z_P = 8$ $E_{con} = 1.5 \text{ V}$
wave wound $\rightarrow Z_a = 2$

$$\frac{Z}{Z_a} = \frac{400}{2} = 200 \text{ Cond / Path}$$

$$E_a = \frac{Z}{Z_a} \times E_{con} = 200 \times 1.5$$

$$E_a = 300 \text{ V}$$

2) $Z = 600$ $Z_P = 6$ $E_a = 200 \text{ V}$
 $E_{con} = ?$ (lap wound $\rightarrow Z_P = Z_a = 6$)

$$E_a = \frac{Z}{Z_a} \times E_{con}$$
$$E_{con} = \frac{200}{\left(\frac{600}{6}\right)}$$

$$E_{con} = 2 \text{ V}$$

3)

$$Z = 600$$

$$2P = 4$$

$$R_{con} = 5 \times 10^{-4}$$

$$(aP \text{ wound}) \rightarrow 2P = 2a = 4$$

$$R_a = ?$$

$$\frac{Z}{2a} = \frac{600}{4} = 150 \text{ con/path}$$

$$R_{path} = 150 \times 5 \times 10^{-4} = 0.075$$

$$R_a = \frac{R_{path}}{4}$$

$$\therefore (R_a = 0.01875 \Omega)$$

4)

$$2P = 8$$

$$\text{wave wound} \rightarrow 2a = 2$$

$$I_a = 50 \text{ A}$$

$$I_c = ?$$

$$I_c = \frac{I_a}{2a} = \frac{50}{2}$$

$$\therefore (I_c = 25 \text{ A})$$

5)

false

$(aP \rightarrow \text{high current low voltage})$

7) C

$$y_c = 1$$

8) C

$$y = y_c = y_b + y_f = \frac{c \pm 1}{P}$$

$\oplus \rightarrow$ Progressive^P

$\ominus \rightarrow$ Retrogressive

9) C

$$z_P = 6$$

$$I_a P \rightarrow z_a = z_P = 6$$

$$S = 120$$

$$\frac{z}{S} = 8$$

$$\therefore Z = 960 \text{ (md)}$$

10)

$$\phi = ?$$

$$N = 500 \text{ rpm}$$

$$E_a = 240 \text{ V}$$

$$E_a = K_a \phi \omega_m = \frac{z_P z}{2\pi z_a} \phi \frac{2\pi N}{60}$$

$$\therefore \phi = 0.03 \text{ wb}$$

11)

$$I_a = 30 \text{ A}$$

$$P_{dev} = ?$$

$$P_{dev} = E_a \times I_a = 240 \times 30$$

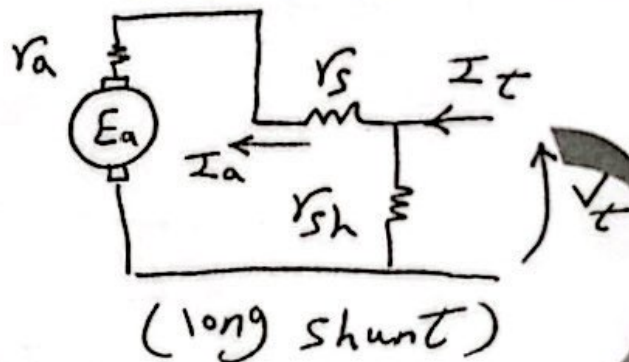
$$\therefore P_{dev} = 7200 \text{ W}$$

$$12) f_a = \frac{1}{8} \frac{\tau I_a}{aP}$$

$$f_a = \frac{1}{8} * \frac{960 * 30}{3 * 3}$$

$$\therefore f_a = 400 \text{ A.T/Pole}$$

13) d



14) d

15) b

16) b

$$17) E_{co} = \frac{2PN\Phi N}{60} = \frac{2PN}{60} B A_{pole}$$

$$E_{con} = \frac{2PN}{60} B \frac{2\pi r \cdot l}{2P}$$

$$E_{con} = 31.42 \text{ V}$$

$$f_a = 6000 \text{ A.T/Pole}$$

$$2P = 6$$

$$\theta_m = 30^\circ$$

$$f_d = f_a \times \frac{2\theta_e}{\pi} = 6000 \times \frac{2 \times \left(\frac{30\pi}{180} \right) \times 3}{\pi}$$

$$\therefore f_d = 6000 \text{ A.T/Pole}$$

$$f_g = f_a - f_d = 0 \text{ A.T/Pole}$$

$$f_a = 6000 \text{ A.T/Pole}$$

$$2P = 6$$

$$f_d = ?$$

$$\theta_m = 2^\circ$$

$$19) f_d = f_a \times \frac{2\theta_e}{\pi} = 6000 \times \frac{2 \times \left(\frac{2\pi}{180} \right) \times 3}{\pi}$$

$$\therefore f_d = 400 \text{ A.T/Pole}$$

$$20) f_g = f_a - f_d = 5600 \text{ A.T/Pole}$$

$$21) \frac{1}{2} \tau_c = \frac{\alpha}{\rho} \times \frac{\tau \Sigma a}{4aP} = 0.75 \times \frac{288}{4 \times 1 \times 3}$$

$$\text{wave} \rightarrow a = 1, P = 3$$

$$\therefore \tau_c = 18$$

$$22) R_f = \text{slope} = \frac{165}{0.75} = 220 \Omega$$

$$23) \frac{E}{N} = \frac{E_c}{N_c} \rightarrow \text{at 4th Point}$$

$$\frac{96}{1000} = \frac{E_c}{N_c}$$

$$220 = \frac{E_c}{0.3}$$

$$E_c = 66 \text{ V}$$

$$\therefore N_c = 687.5 \text{ rpm}$$

$$24) R_c = \text{slope at 2nd Point}$$

$$R_{fc} = \frac{32}{0.1} = 320 \Omega$$

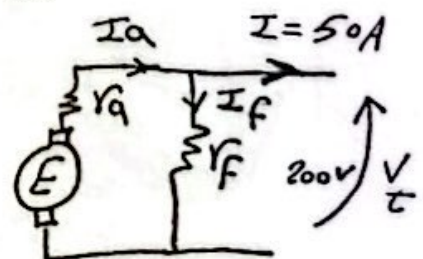
25) false

26))

$$27) P_o = 50 \times 200 = 10,000 \text{ W}$$

$$I_f = \frac{200}{100} = 2 \text{ A}$$

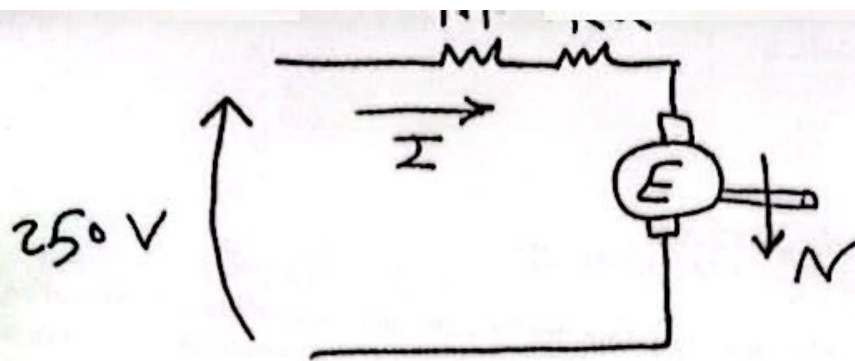
$$I_a = 52 \text{ A}$$



$$P_{dev} = E I_a \quad E = 200 + 50 \times 0.2$$

$$P_{dev} = 10920 \text{ W} \quad P_{in} = P_{dev} + P_{rot} = 11320 \text{ W}$$

$$\therefore \eta = 88.3\%$$



$$I = 50 \text{ A} \rightarrow E_1 = 96 \text{ V} \text{ at } N_1 = 1200$$

From circuit $E_2 = 250 - 50 \times 0.08 = 246 \text{ V}$

$$E \propto I_f N$$

$$\frac{E_1}{E_2} = \frac{N_1}{N_2} \quad \frac{96}{246} = \frac{1200}{N_2}$$

$\therefore$

$$N_2 = 3075 \text{ RPM}$$

29) $P_{dev} = E I_a = 246 \times 50 = T_d \times \omega$

$$\frac{2\pi N_2}{60}$$

$$T_d = 38.2 \text{ N.m}$$

30)